National University of Computer & Emerging Sciences, Karachi
Fall-2023 FAST School of Computing
Mid-Term - II Exam

Answer the following.

a. Rolle's Theorem is used to find the zeros of a function. (True/**False**)

b. $\lim\limits_{x\to 0^+} sinxlnx$

I). **0** II) $+\infty$ III). -1 IV). $\frac{\pi}{2}$ V) 2

c. For what value of x the function $f(x) = x^3 - 8$ on $[3,7]$ satisfies the mean value theorem.

I). 4.509 II). 3.512 III). 8.888 IV). **5.132** V). 6.285

d. $\lim\limits_{t\to 0} \frac{te^t}{1-e^t}$

I). 0 II) $-\infty$ III). **-1** IV). $\frac{-1}{e}$ V) $-e$

e. The rational function $f(x) = \frac{3-x^2}{x^3}$, has

I). a stationary point at $x = 1$

II). a stationary point at $x = -1$

III). **two stationary points at approximately $x = -1.723$ and $x = 1.723$**

IV). three stationary points at approximately $x = 0$, $x = -1.723$ and $x = 1.723$

V). no stationary points

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a. Use the definition of area under the curve with x_k^* as the midpoint of the subinterval to find the area under the curve $y = 4x - x^2$ over the interval $[0, 4]$.

8 $y = 4x - x^2$ $[0,4]$.

$\Delta x = \frac{b-a}{n} = \frac{4}{n}$

$x_k^* = a + (k - \frac{1}{2})\Delta x$
$= (k - \frac{1}{2})\frac{4}{n}$

$f(x_k^*) = 4(k-\frac{1}{2})\frac{4}{n} - [(k-\frac{1}{2})\frac{4}{n}]^2$

$= \frac{16}{n}(k-\frac{1}{2}) - \frac{16}{n^2}(k-\frac{1}{2})^2$

$= \frac{16}{} [\frac{k}{n} - \frac{1}{2n} - \frac{1}{n^2}(k^2 - k + \frac{1}{4})]$

$\sum f(x_k^*).\Delta x = 16\times4[\frac{1}{n^2}\sum k - \frac{1}{2n^2}\sum 1 - \frac{1}{n^3}\sum k^2 + \frac{1}{n^3}\sum k - \frac{1}{4n^3}\sum 1]$

$A = \lim\limits_{n\to+\infty} \sum f(x_k^*).\Delta x$

$= 16[\lim\limits_{n\to+\infty} \overbrace{\frac{1}{n^2}\sum k}^{1/2} - \frac{1}{2}.\lim\limits_{n\to+\infty} \overbrace{\frac{1}{n^2}\sum 1}^{1\times 0} - \lim\limits_{n\to+\infty} \overbrace{\frac{1}{n^3}\sum k^2}^{1/3} +$
$\lim\limits_{n\to+\infty} \overbrace{\frac{1}{n^3}\sum k}^{0\times 1/2} - \frac{1}{4}\lim\limits_{n\to+\infty} \overbrace{\frac{1}{n^3}\sum 1}^{0\times 1}] \times 4$

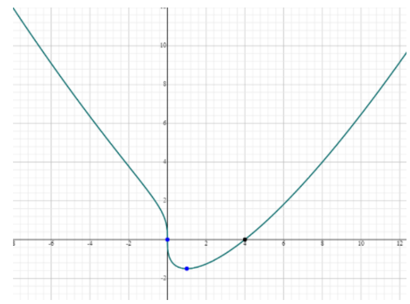
$= 16[\frac{1}{2} - \frac{1}{3}]4$

$\boxed{A = \frac{32}{3}.}$

b. If $f(x) = \frac{1}{2}x^{\frac{4}{3}} - 2x^{\frac{1}{3}}$, Find

   I. All critical points
   II. Intervals in which function is increasing and decreasing.
   III. Relative extrema

Solution:



(graph is not required from students)

$$f'(x) = \frac{2}{3} * \frac{x-1}{x^{2/3}}$$

Critical points:

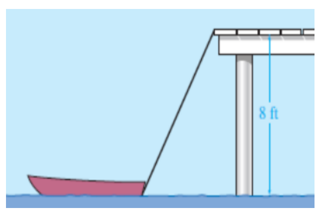
$x = 0$, (Non-stationary) $x = 1$ (stationary)

Intervals of decreasing $(-\infty, 0), (0,1)$

Intervals of increasing $(1, +\infty)$

Relative minima at $x = 1$, Relative maxima does not exists.

c. A pulley is on the edge of a dock, 8 *ft* above the water level. (See the figure below.) A rope is being used to pull in a boat. The rope is attached to the boat at water level. The rope is being pulled in at the rate of 1 *ft* per second. Find the rate at which the boat is approaching the dock at the instant the boat is 4 *ft* from the dock.



Solution:

We will describe the variables below

Step 1: We are told the rope is pulled in at a rate of 1 foot per second. This is denoted by $\frac{dz}{dt} = -1$. The negative sign just denotes that the line is being shortened. We are asked to find $\frac{dx}{dt}$ when $x = 4$.

Step 2: The variables are related through the Pythagorean Theorem with

$$z^2 = x^2 + 8^2$$

Step 3: Differentiating gives us

$$2z\frac{dz}{dt} = 2x\frac{dx}{dt}$$

Step 4: Recall we are given $\frac{dz}{dt} = -1$. When $x = 4$ we have

$$z^2 = 4^2 + 8^2 z = \sqrt{80}$$

Plugging these in the equation from the previous step we have

$$2z\frac{dz}{dt} = 2x\frac{dx}{dt} 2(\sqrt{80})(-1) = 2(4)\frac{dx}{dt} \frac{dx}{dt} = \frac{-\sqrt{80}}{4}\frac{dx}{dt} = -2.2361 \text{ feet per second.}$$

Find the integral of the following.

a. $\int_1^4 \frac{\ln(x)}{x^{\frac{1}{2}}} dx$

We integrate by parts. We have

$$\begin{aligned}
\int_1^4 x^{-1/2} \ln(x)\ dx &= \left[\frac{x^{1/2}}{(1/2)} \ln(x)\right]_1^4 - \int_1^4 \frac{x^{1/2}}{(1/2)} . \frac{1}{x}\ dx \\
&= 2\left[\sqrt{4}\ln(4) - \ln(1)\right] - 2\int_1^4 x^{-1/2}\ dx \\
&= 4\ln(4) - 2\left[\frac{x^{1/2}}{(1/2)}\right]_1^4 \\
&= 4\ln(4) - 4\left[\sqrt{4} - 1\right] \\
&= 4(\ln(4) - 1).
\end{aligned}$$

b. $\int \frac{\sqrt{1+4x^2}}{x} dx$

26. $2x = \tan\theta,\ 2\,dx = \sec^2\theta\,d\theta,\ \displaystyle\int \sec^2\theta\csc\theta\,d\theta = \int(\sec\theta\tan\theta + \csc\theta)\,d\theta = \sec\theta - \ln|\csc\theta + \cot\theta| + C = \sqrt{1+4x^2} - \ln\left|\frac{\sqrt{1+4x^2}}{2x} + \frac{1}{2x}\right| + C.$